

P2 Bosonic Coleman–Weinberg Matching Hessian for the Hierarchical $P54PQ$ -v2 Vacuum

Route-F four-dimensional mainline

2 September 2026

Abstract

We compute the bosonic one-loop zero-momentum matching curvature on the real four-plane of the tuned Standard-Model Higgs doublet. The declared scheme is background-field Landau gauge and $\overline{\text{MS}}$ at $\mu_U = g_U \omega$. A hard/soft spectral split retains all 290 massive physical scalar directions and 33 massive vector directions while leaving the 38 scalar and 12 vector infrared degrees of freedom in the lower-energy theory. A matrix-function Fréchet formula differentiates the complete field-dependent 328×328 scalar Hessian without subtracting nearly equal one-loop potentials. At the fixed point $\sigma/\omega = 0.1265$, the result is

$$\Pi_B^{\text{hard}}/\omega^2 = -0.0202366510950 \mathbf{1}_4, \quad \delta\xi_{02,B} = -0.0202367022071.$$

This is a scheme-dependent matching coefficient, not a gauge-independent pole mass. The unfit heavy-fermion projection is retained explicitly as $\eta_Y(\mu)$, so the renormalization condition is $\delta\xi_{02} = [\kappa_B + \eta_Y]/w_{10}$.

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1 Background and the precise claim

The input is the previously verified hierarchical stationary point

$$\frac{\sigma}{\omega} = 0.1265, \quad \frac{v_s}{\omega} = 0.25, \quad g_U = 0.562602889985, \quad (1)$$

with tree parameter $\xi_{02} = 0.147371094023$. Its complete real scalar Hessian has dimension

$$54 + 2(126) + 2(10) + 2 = 328.$$

There are 38 zero directions: 33 gauge-orbit directions, one physical PQ Goldstone, and a real four-plane L carrying one $(\mathbf{1}, \mathbf{2}, |Y| = 1/2)$ multiplet. There are no negative modes. The next doublet has

$$\Delta_D = 0.0305570302561 \omega^2. \quad (2)$$

The calculation below determines the heavy-boson part of the one-loop zero-momentum matching matrix on L . It does *not* determine a gauge-independent pole mass. The latter additionally requires the momentum-dependent self-energy, wave-function matching, a consistent counterterm prescription, and the fitted heavy-Yukawa matrices. This distinction is compulsory because an effective potential away from a stationary point is gauge dependent; the Nielsen identity does not turn an arbitrary Hessian entry into an observable [2].

2 Gauge, ghost, and infrared prescription

We use background-field R_ξ gauge at

$$\xi_{\text{gf}} = 0 \quad (\text{Landau gauge}) \quad (3)$$

and $\overline{\text{MS}}$. At constant background the hard bosonic functional is

$$V_{1,B}^{\text{hard}}(x; \mu) = \frac{1}{64\pi^2} \left[\text{tr}_{290} f_s(M_s^2(x)) + 3 \text{tr}_{33} f_v(M_v^2(x)) \right], \quad (4)$$

$$f_s(z) = z^2 \left(\log \frac{z}{\mu^2} - \frac{3}{2} \right), \quad f_v(z) = z^2 \left(\log \frac{z}{\mu^2} - \frac{5}{6} \right). \quad (5)$$

The vector coefficient three and the $5/6$ finite constant are the usual Landau- $\overline{\text{MS}}$ vector result [1, 3]. The Faddeev–Popov eigenvalues are $m_{\text{gh}}^2 = \xi_{\text{gf}} M_v^2 = 0$; hence there is no hard ghost term in this declared scheme.

The subscripts on the traces are a Wilsonian spectral split. The 290 positive scalar eigenvalues are retained; the 38 zero modes are assigned to the infrared theory. Similarly, the 33 positive vector eigenvalues are retained and the 12 unbroken vectors are infrared fields. This split is locally unambiguous because

$$\lambda_{s,\text{min}}^{\text{hard}} = 0.0135204700328 \omega^2, \quad \lambda_{v,\text{min}}^{\text{hard}} = 0.01600225 g_U^2 \omega^2, \quad (6)$$

whereas all soft eigenvalues vanish at the matching background. Hard–soft mixing is retained in the divided-difference term below; “remove the soft trace” therefore does not mean projecting every derivative onto the hard subspace.

3 First-principles second variation

Let z^a , $a = 1, \dots, 4$, be orthonormal canonical coordinates on L . For either bosonic mass operator, write

$$M^2(z) = M_0^2 + z^a A_a + \frac{1}{2} z^a z^b B_{ab} + O(|z|^3). \quad (7)$$

If $M_0^2 = U \text{diag}(\lambda_i) U^T$, define all matrices below in this eigenbasis. Differentiating $\text{tr} f(M^2)$ once gives

$$\partial_a \text{tr} f(M^2) = \text{tr} [f'(M^2) \partial_a M^2]. \quad (8)$$

The Fréchet derivative of f' has components

$$[Df'(M_0^2)[A_b]]_{ij} = f'^{[1]}(\lambda_i, \lambda_j)(A_b)_{ij}, \quad (9)$$

where

$$f'^{[1]}(x, y) = \begin{cases} \frac{f'(x) - f'(y)}{x - y}, & x \neq y, \\ f''(x), & x = y. \end{cases} \quad (10)$$

Therefore the exact second variation is

$$\partial_a \partial_b \text{tr} f(M^2) = \sum_i f'(\lambda_i)(B_{ab})_{ii} + \sum_{i,j} f'^{[1]}(\lambda_i, \lambda_j)(A_a)_{ij}(A_b)_{ji}. \quad (11)$$

For a soft eigenvalue we set $f' = f'' = 0$, the derivative of the locally flat infrared subtraction. Equation (11) still includes the hard-soft terms because their divided differences are nonzero.

For $f_c(z) = z^2[\log(z/\mu^2) - c]$, the derivatives used in the code are

$$\begin{aligned} f'_c(z) &= z \left[2 \log \frac{z}{\mu^2} - 2c + 1 \right], \\ f''_c(z) &= 2 \log \frac{z}{\mu^2} - 2c + 3. \end{aligned} \quad (12)$$

The scalar mass operator is not reconstructed from a tabulated spectrum: it is the automatic second derivative of the full tensor potential,

$$M_s^2(x) = \frac{\partial^2 V_{\text{tree}}}{\partial x^I \partial x^J}. \quad (13)$$

Because the action is quartic, $M_s^2(x)$ is exactly quadratic in x . Thus centered evaluations at $\epsilon/\omega = 0.02$ determine A_a and B_{aa} , and a combined displacement determines B_{01} . Repeating one direction at $\epsilon/\omega = 0.01$ gives relative residuals

$$r_A = 2.32 \times 10^{-14}, \quad r_B = 2.15 \times 10^{-11}. \quad (14)$$

For vectors the derivation is analytic. If $O_A(x) = T_A x$ is the gauge-orbit matrix, then

$$M_v^2(x) = g_U^2 O(x)^T O(x), \quad (15)$$

so, with $O_a = T q_a$,

$$\begin{aligned} A_a^{(v)} &= g_U^2 (O_a^T O_0 + O_0^T O_a), \\ B_{ab}^{(v)} &= g_U^2 (O_a^T O_b + O_b^T O_a). \end{aligned} \quad (16)$$

4 Light-doublet basis and symmetry reduction

Let Z be the tree zero-eigenspace and let S contain the 33 gauge directions plus the physical PQ direction. The numerical light projector is obtained from

$$L = \text{range}[(1 - SS^+)Z]. \quad (17)$$

Its rank is four and its Casimirs are $C_3 = 0$, $C_2 = 3/4$, and $Y^2 = 1/4$. The hard spectral traces commute with the unbroken Standard Model. The realification of this charged complex doublet has a unique invariant symmetric bilinear form. Consequently

$$\Pi_B^{\text{hard}}|_L = \kappa_B \mathbf{1}_4. \quad (18)$$

This is a symmetry statement, not a numerical fit. Numerically, two independent diagonal directions agree to 2.29×10^{-13} , the tested mixed component is $7.70 \times 10^{-14} \omega^2$, and the complete vector block has isotropy residual 2.94×10^{-18} .

5 Numerical result and scale dependence

At $\mu_U = g_U \omega$, Eq. (11) gives

$$\begin{aligned} \frac{\Pi_s^{\text{hard}}}{\omega^2} &= -0.0182525705951 \mathbf{1}_4, \\ \frac{\Pi_v^{\text{hard}}}{\omega^2} &= -0.00198408049986 \mathbf{1}_4, \\ \boxed{\frac{\Pi_B^{\text{hard}}}{\omega^2} &= -0.0202366510950 \mathbf{1}_4} \end{aligned} \quad (19)$$

The off-diagonal entries of the directly assembled vector matrix are below $1.1 \times 10^{-18} \omega^2$. The four equal components in Eq. (19) are fixed by Eq. (18); only two diagonal directions and one mixed direction need expensive scalar evaluations.

The coefficient is scale dependent, as a matching coefficient must be:

μ/μ_U	κ_s/ω^2	κ_v/ω^2	κ_B/ω^2	$\delta\xi_{02,B}$
0.5	0.0107814599	0.0060371453	0.0168186052	0.0168186477
1.0	-0.0182525706	-0.0019840805	-0.0202366511	-0.0202367022
2.0	-0.0472866011	-0.0100053063	-0.0572919074	-0.0572920521

The large finite-scale movement is not an uncertainty band. It is cancelled order by order by the running renormalized parameter and by the scale dependence of the fermionic matching contribution. Quoting only the middle number as a physical prediction would therefore be incorrect.

6 Retuning and the heavy-Yukawa nuisance

The normalized light mode has

$$w_{10} = \langle h | P_{10} | h \rangle = 0.999997474290. \quad (20)$$

Since the tree Hessian responds to $\delta\xi_{02}$ as $-\delta\xi_{02}P_{10}$, the bosonic counterterm required to keep the doublet massless at the matching surface is

$$\delta\xi_{02,B}(\mu_U) = \frac{\kappa_B(\mu_U)/\omega^2}{w_{10}} = -0.0202367022071. \quad (21)$$

The heavy-fermion contribution is deliberately not guessed. Define the matching nuisance

$$\eta_Y(\mu) = \frac{1}{\omega^2} \langle h | \Pi_{\text{heavy-Y}}(0; \mu) | h \rangle. \quad (22)$$

The complete matching relation carried into the global flavor analysis is then

$$\boxed{\delta\xi_{02}(\mu) = \frac{\kappa_B(\mu)/\omega^2 + \eta_Y(\mu)}{w_{10}}}. \quad (23)$$

No value or prior for η_Y is fabricated here. The independent one-light-doublet safety gate remains

$$\|Q [\Pi_B + \Pi_Y - \delta\xi_{02}P_{10}] Q\| < \Delta_D, \quad Q = 1 - |h\rangle\langle h|, \quad (24)$$

with Δ_D from Eq. (2). A global heavy-Yukawa fit must evaluate Eq. (24); the present calculation closes the bosonic matching coefficient, not that later flavor-dependent norm bound.

7 Reproducibility and promotion boundary

The machine calculation is

```
route_f/code/verify_p54_p2_bosonic_cw.py
route_f/output/p54_p2_bosonic_cw.json
route_f/output/p54_p2_bosonic_cw.md
```

It evaluates eight complete field-dependent scalar Hessians and passes 14 of 14 physics/numerical checks. The soft-soft first-derivative norms are 3.88×10^{-15} and 3.08×10^{-15} , closing the declared IR split. An independent random-matrix regression of Eq. (11) against a direct four-point finite difference has relative residual 2.47×10^{-9} .

The P2 bosonic matching task is therefore closed in the declared scheme. The full pole mass remains unpredicted, and η_Y is now an explicit matching nuisance rather than a hidden assumption. This is the appropriate boundary at which P3 may begin a global flavor/seesaw fit: P3 must fit or profile η_Y , propagate it through Eq. (23), and test Eq. (24).

References

- [1] S. Coleman and E. Weinberg, “Radiative Corrections as the Origin of Spontaneous Symmetry Breaking,” *Phys. Rev. D* **7** (1973) 1888.
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- [3] S. P. Martin, “Two-loop effective potential for a general renormalizable theory and softly broken supersymmetry,” *Phys. Rev. D* **65** (2002) 116003, arXiv:hep-ph/0111209.